



1.04 Loading

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps in loading the Vacuum Truck

2. Scope

This procedure is intended for use by all Silverpoint employees .

3. Hazard Assessment & Identification

- Slips , Trips and Falls
- Physical Injury - Ergonomics
- Pinch Points
- Lifting , Twisting , Pulling , Stretching
- Personal Injury
- Equipment Failure /Stress
- Excessive Noise
- Release to Environment
- Physical Stress
- Mental Stress
- Hidden Objects Under Drilling Fluid
- Excessive Hose Movement While Loading and Unloading

4. EHS Requirements

- Fire Retardant Coveralls .
- Steel Toed Boots .
- Safety Glasses .
- Hardhat
- Gloves
- Hearing Protection



5. References

- Risk Management Manual (RMM) 5.2
- Silverpoint Safety handbook
- Alberta OHS Regulations
- Task list
- MED HAT THA 1.04

6. Procedure

- 6.1. **Loading** - After Vacuum Truck is in position , then get out and place the chock blocks and check that the blower switch is off, and all rear valves are closed . Connect the hose to the load valve on back of truck (SOP 1.02). When loading Vacuum Truck, the four-way valve handle should be pointed to Vacuum ; this should be clearly labeled if not labeled then make arrangements to label it. If you're not sure don't hesitate to call the shop .

Loading from the Rig

- 6.1.1 . When backing into position , it is always a good idea to find out which tank the Derrick hand would like you to suck out. When you are backing up, make sure that the area is clear, and **it is company policy to use a Spotter** (SOP 2.01) that way they can watch for anyone that might cross your path or any obstructions in your path. Try and park so that you can see your gauge , so you know when you are getting full .
- 6.1.2 . When you are stringing out your hoses , make sure that you have enough length , so that if you need to do an extra tank you don't have to move the truck or find another hose (just to save you some time) . If you are working with a rig that has hook-ups on the outside of their tanks , make sure that you hook up your hose properly . If you have to go over top of the tanks , try to go through the railing instead of over the top (your hose will kink off) . It helps if you tie your hose to the railing as well , to stop the hose from kicking and maybe hurting someone .
- 6.1.3 . Refer to Connecting and disconnecting hoses SOP 1.02, 6.1.1 .
- LOADING
- 6.1.3.1 . Determine where you will be loading from: Tank, Sump ?
- 6.1.3.2 . Determine length and how many hoses you will need to load .
- 6.1.3.3 . Connect first hose to truck (make sure Vac Unit is in Vacuum Mode)
- 6.1.3.4 . Connect second hose and more if required .
- 6.1.3.5 . Your last connection would be the coupler on the tank or the placement of the hose into a tank or sump .

UNLOADING (Refer to Transferring Fluids SOP 1.09 6.1)



- 6.1.3.6 . Determine where you will be unloading to: Tank, Sump ?
- 6.1.3.7 . Determine length and how many hoses you will need to unload .
- 6.1.3.8 . Make sure the unit is in Pressure mode .
- 6.1.3.9 . Connect first hose to truck .
- 6.1.3.10 . Connect second hose and more if required .
- 6.1.3.11 . Your last connection would be the coupler on the tank or the placement of the hose into a tank or sump .

VAC Unit Operation : For Vacuum unit operation during unloading Refer to SOP 1.09

- 6.1.4 . **Whip Checks are required where Vac Tank pressure will be or is used to unload Product.**
- 6.1.5 . When the vacuum hose draws air while loading fluid or solids it can jump /kick very severely , always leave room for hose movement and warn others working in the area , secure hose if needed
- 6.1.6 . Every rig setup is about the same . The different compartments are usually the same , they are:
 - 6.1.6.1 . **Shaker and Shaker Box** – the shaker is a piece of equipment that vibrates and screens the shale out of the mud and flock . The shaker box is attached on the end of the shaker . The shale travels along the screens and then down a ramp and into the shale compartment of the flock tank . The mud and flock water falls through the screens and carries on .
 - 6.1.6.2 . **Sand Trap** – as the mud or flock water comes back out of the well , it passes over the SAND TRAP , the sand drops down and the fluid keeps going into the rig tank . This helps keep the weight and viscosity down as it goes back into the rig tank (also helps keep the tanks cleaner) . You will need to clean this at the end of surface and main hole . Some derrick hands might get you to clean it while they are flocking as well . If you are not sure where the sand trap is , ASK .
 - 6.1.6.3 . **Shaker Tank** – this compartment is located under the shaker itself (hence the name) . Normally they only use this tank if they need it for flocking . It will need to be cleaned at the end of the hole as well , if used . If they have trouble they can open their bypass , and it will dump fluid and shale into this tank .
 - 6.1.6.4 . **Settling Tank** – the settling tank has a big vessel in it that is used in case they hit a gas pocket while drilling . If they don't use this tank for flocking , they will have a couple of cubes of water in it . This is so the gas goes through the water before going down the flare line ; it is easier to control this way . This will need to be cleaned at the end of the hole as well .
 - 6.1.6.5 . **Mixing Tank** – this tank is where they fine tune their mud or add chemicals for when they are flocking . It is the main tank that they pump fluid down the well with . When doing surface , this is



usually the only tank that they are using . It will need to be cleaned at the end of surface and main hole .

6.1.6.6 . **Pill Tank** – this tank is used to mix up a very thick batch of mud , for when they are losing circulation so that can pump it down the hole in a hurry to get the well under control . Once in a while you will need to clean this tank .

6.1.6.7 . **Trip Tank** – this is usually located close to the shaker . This tank has a submersible pump in it, that they attach a 2” line to, for filling the casing , when running casing . Once in a while they will get you to clean it .

6.1.6.8 . **Chemical Barrel** – this is a 45-gallon drum that they have modified with a lid and a mixing motor . They will mix caustic with water for when they are flocking , if the derrick hand forgets to wash it out, he might ask you to suck it out . Be very careful when doing this , as he has used harmful chemicals that could burn your skin . Try and use water to wash out when you are sucking it out .

6.1.7 . When you are loading from the rig do not assume that you are taking a full load , they might only want you to take a few cubes . Make sure that you talk to you derrick hand and have good communication with him , to avoid any confusion .

6.2. Loading from Flock Tanks

6.2.1 . When you are loading from the flock tanks , you can either back in or pull along side the tanks . If you pull along side the tanks , you wouldn’t have to back in anywhere and you could still see the tank gauge on the truck good . If you are going to back up to the tank , **make sure that you have a spotter to guide you back** .

6.2.2 . Flocking is the process of drilling with water . By mixing calcium , flock gel and water , it allows the solids that come up from the hole being drilled , over the shaker and into the flock tank , to settle to the bottom of the flock tanks , leaving in return clear and clean water on top . As it comes into the flock tank and travels to each compartment , the solids settle to the bottom as the good water moves on to the last compartment , and then the water is pumped back into the rig tank and down the hole while drilling . This process is also known as **Fast Holing** .

6.2.3 . One of the best tools that we have on the trucks is called a **“Flock Rod”** or **“T bar”** . This is a 1” piece of pipe with a handle on the end to hang onto and a special cam lock end so you can attach a hose to it, connected to the 1” pipe with a 1” hammer union . Even if you are not flocking you can use this tool to suck up shale , it just gives you better control when doing this . Air travels through the flock rod when sucking so you won’t plug your hose .

6.2.4 . **Safety Alert : This job is repetitive so take a break every ten minutes for a few seconds and stretch out muscles . Move body position frequently . Switch out with other worker if possible , at frequent intervals . Recognize when you are over stressing your muscles . Do not wait for muscle strains , pulls before changing position . Do not use long strokes when overhead use of the flock rod . Avoid overhead strokes if possible .**

6.2.5 . While they are flocking it is very important that you have good communication with your derrick hand , only load when told , and make sure that you know which compartment you are supposed to load from . If you are



unsure or didn't understand , ask again .

- 6.2.6. Standard operating procedure will have the operator manning the controls on the operation of the Vac truck, the other worker will be cleaning the tank .
- 6.2.7. The worker should do as much of the work as they can from outside of the flock tank
- 6.2.8. A worker cleaning the flock tanks ,should start at the shale end ,that way the farther back you go the cleaner the fluid and helps keep your truck cleaner . Make sure that you squeegee the compartments out as well ,this procedure can only be done with a monitor present . If you are done loading make sure that your hose is not connected to flock tanks , with the other end on the ground . This may cause the tank to siphon onto the ground .
- 6.2.9. You may need to go inside the flock tank to perform your task . Access in or out of the tank has to be performed with caution . Perform a hazard assessment prior to entry ,determine your access point . Use three points of contact always facing the tanks walls . Refer to SOP # 4.01

6.3. Cleaning Tanks, Boiler and Pump

- 6.3.1. **Safety Alert:** Always check for equipment or loose objects in the tanks before walking thru the tank ,rig crew will often throw hoses , pipe , rig parts or wood into the tank when they move the rig . If they don't clean the tank out before they pump mud into it these pieces are hidden from your view while you are trying to clean the tank . Always confirm that the bottom of the tank is clear by loading as much from the tank as possible before entering and use a shovel or a squeegee to check the floor of the tank for hidden objects . **Failure to check the tank properly has resulted in serious injury in the past**
- 6.3.2. Always have good communication with your derrick hand ,ask him where he would like to start cleaning tanks ,which compartment . Especially after surface , sometimes they want you to start with a certain tank so they can start mixing their mud sooner . Always take care of the rig before you clean the flock tanks ,because that way the rig can move while you are finishing the flock tanks (usually the last thing to move) and you can transfer a load .
- 6.3.3. Normally when cleaning the rig tanks ,the derrick hand is expected to go in and clean ,if you are going to go in and clean with him ,make sure that a confined space is done (either theirs or ours) and the tank has been tested . You should have a standby person as well in case one of you goes down . If you do not go in ,be there for him and get what ever he needs (shovel , squeegee , water hose or anything else) and let him know from time to time how full your tank is getting . Depending on the work sites that you are working at ,you may be required to work alone . If you are alone or you are working with another , complete a FLHA and when required take into consideration the Work Alone procedure . Also , a flock tank is not clean until it is squeegeed out .
- 6.3.4. In the winter the rigs have boilers ; they will get you to suck out the boiler after they have blown it down . This usually happens if a project is done ,and they are shutting down . While they are drilling most of them have a container that they blow out their lines into from time to time (if they have water in their lines) . You will need to suck out this container once in a while , so it doesn't freeze . Be very careful when working around the boiler ,there are **High Pressure Lines** and **very hot** steam in these lines .



- 6.3.5. Once in a while the derrick hand will get you to suck out the screen for the pump . This is located in the pump shack and only takes a couple of seconds to do. Sometimes this is referred to as the **"Bear Trap"**. If there is anything that has to come apart , he should be the one doing it, because it is their equipment and might be under pressure . **A 2" hose works really well for this.**
- 6.3.6. Most of the time when you are working in the shallow gas areas , they would like you to transfer a load of mud at the end of the hole. This allows the rig to get drilling surface a lot faster , saves water and helps the water haulers out as well . If you are going to do this refer to SOP 1.09 to properly do this procedure .
- 6.3.7. If you have any questions about the rig, drilling procedure or anything else, ask. Most of the guys that we work with are more then willing to explain things to you as long as it doesn't interfere with their work . Ask your supervisor as well , if they don't know they will find out for you .
- 6.3.8. Do not lay your hose down while still connected to the tank as this will cause the siphon effect and drain the contents of the flock tank onto the ground creating a spill to clean and report.**

6.4. Loading from a 400 BBL

Before hooking up to any 400 bbl tank check with the site supervisor if the vent is open , if they are unsure , it is always best to suggest that someone go up and check .

Do Not climb the tank yourself to check the vent as you do not have the proper training or PPE to safely perform the job .

- 6.5. When you are working on deeper holes , it is your responsibility to suck out the cellar and ditches .** The best way to keep on top of this is to fit it into your daily routine , in the morning suck out these places , then see if the Derrick hand needs anything done . If there is rain, they might get you to suck up the puddles , so the lease dries out faster, which is good for everybody . You might have to suck out the ditches and cellar in the morning and then in the evening as well . Check with your Derrick hand to see if he needs the extra water you sucked up, if not then spread it, you might want to put it into an empty flock compartment till you have a full load to spread .

Remember : if you are in doubt about what to do, call your supervisor